

Evaluating a Next-Generation Long-Context Model for Financial Document Reasoning: A Comparison with Retrieval-Augmented Prompting for Cross-Section Inconsistency Detection in SEC 10-K Filings

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Declaration

This dissertation is submitted in part fulfilment of the requirements for the degree of MSc Advanced Computer Science with Artificial Intelligence of the University of Strathclyde.

I declare that this dissertation embodies the results of my own work and that it has been composed by myself.

Following normal academic conventions, I have made due acknowledgement to the work of others.

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Abstract

This dissertation investigates evidence-grounded reasoning over lengthy SEC 10-K filings, where relevant information may be distributed across Business, Risk Factors, MD&A, financial statements and notes. The research addresses a specific gap in the literature: it remains unclear whether expanded context capacity improves cross-section inconsistency detection compared with RAG-based context delivery when the selected model, filing corpus, claim-cases, task instructions, label definitions, output schema and scoring procedure are held constant.

The aim of the research was to evaluate whether the Long_Context strategy, using the selected GPT-5.5 API model, provided measurable advantages over the RAG-based strategy for SEC 10-K cross-section inconsistency detection. A Type 5 experimental design was used, based on a controlled same-model comparison. Under the Long_Context strategy, the model received the full cleaned filing. Under the RAG-based strategy, the same model received retrieved chunks from the same filing corpus. The experiment used six official SEC Form 10-K filings, 48 locked claim-cases and 96 completed strategy-level calls, with claim-cases balanced across Supported, Contradicted and Insufficient Evidence.

The findings did not support an overall Long_Context advantage. Long_Context achieved 85.42% accuracy and 84.68% macro F1, while the RAG-based strategy achieved 87.50% accuracy and 87.28% macro F1. The RAG-based strategy therefore classified one additional claim-case correctly and achieved slightly stronger overall classification performance. Operationally, Long_Context used approximately 19.36 times more input tokens, incurred approximately 15.49 times higher telemetry-estimated model-call cost, and produced higher mean and median latency.

The main diagnostic finding was that expanded access to filing evidence did not guarantee better evidence use. Both strategies identified all true Contradicted claim-cases, but most misclassified outputs occurred at the boundary between Insufficient Evidence and Contradicted, especially where related disclosure did not directly establish causal, forward-looking, external or metric-specific assertions. The dissertation concludes that neither the Long_Context strategy nor the RAG-based strategy is sufficient by itself to guarantee accurate evidence-grounded SEC 10-K reasoning. Effective use requires claim-specific evidence integration, explicit handling of Insufficient Evidence, retrieval completeness where retrieval is used, and disciplined label-boundary application.

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Chapter 1: Introduction

This chapter introduces the research context, problem, aim, research questions, scope and expected contribution. In this dissertation, *controlled* means that the selected GPT-5.5 API model and experimental inputs are held constant while only the way filing evidence is supplied to the model is varied. The chapter establishes why this controlled same-model comparison of the *Long_Context* and *RAG-based* strategies is required for evidence-grounded cross-section inconsistency detection in SEC 10-K filings. Section 1.4 defines the governing terminology used consistently throughout the dissertation.

1.1. Research Background and Context

Next-generation large language models (LLMs) support expanded context windows that allow substantially longer documents to be supplied directly within a single prompt. Long-context benchmarks such as *LongBench*, *L-Eval* and *RULER* evaluate performance over extended inputs and effective context use (Bai et al., 2024; An et al., 2024; Hsieh et al., 2024), directly aligning with the *ROUS04* theme’s question of whether massive context windows offer measurable benefits in real-world tasks.

However, expanded input capacity is a computational capability and does not guarantee accurate reasoning. Long-context studies identify positional effects, including the “lost in the middle” effect, whereby relevant evidence may be underused despite being present in the supplied context (Liu et al., 2024; Hsieh et al., 2024).

This limitation matters for SEC 10-K reasoning because claim assessment may require evidence integration across multiple filing sections. SEC Form 10-K filings are official, public and structured, with disclosures distributed across sections including Business, Risk Factors, Management’s Discussion and Analysis, Market Risk, financial statements and notes (U.S. Securities and Exchange Commission, 2026). These characteristics justify the use of SEC 10-K filings for evaluating evidence-grounded cross-section inconsistency detection across long financial documents.

1.2. Research Problem and Gap

Retrieval-Augmented Generation (RAG) provides a retrieval-mediated, prompt-size-reducing context-delivery strategy for knowledge-intensive tasks (Lewis et al., 2020; Gao et al., 2023), making it a relevant comparison strategy for long-document reasoning. However, retrieval omissions, chunk-boundary fragmentation and noisy context can leave the model with an incomplete evidence base (Cuconasu et al., 2024; Barnett et al., 2024; Jin et al., 2024). Direct long-context prompting avoids retrieval preselection but may still fail to integrate distributed

evidence accurately across a long document (Liu et al., 2024; Li, Z. et al., 2024; Li, X. et al., 2025). Because both strategies have distinct evidence-delivery limitations, a fair comparison requires a task-specific, same-model design in which context-delivery strategy is isolated as the independent variable.

SEC 10-K filings provide an appropriate setting for this comparison because they are official, structured and cross-referential documents, combining management commentary, audited financial statements and regulatory disclosures used by investors and analysts to assess corporate performance, risks and prospects (U.S. SEC Office of Investor Education and Advocacy, 2011; CFA Institute, 2026). Their disclosure complexity, readability, consistency and comparability can also affect users' information costs and interpretation (Li, 2008; Loughran and McDonald, 2014; Miller, 2010; Colas and García Osma, c. 2023), supporting their use for this experimental comparison.

Although existing literature addresses evidence-grounded financial reasoning, contradiction detection, hallucination and claim verification (Zhao et al., 2024; Deußer et al., 2023; Huang et al., 2023), it does not resolve whether the Long_Context strategy offers measurable advantages over the RAG-based strategy for SEC 10-K cross-section inconsistency detection when the selected model, filings, claim-cases, label definitions, output schema, scoring procedure and telemetry-capture procedure are held constant.

1.3. Aim and Objectives

The aim of this research is to evaluate whether direct long-context prompting offers measurable advantages over RAG-based prompting for cross-section inconsistency detection in SEC 10-K filings when using the selected GPT-5.5 API model. The comparison is assessed through claim-classification performance, qualitative error and evidence-use patterns, token use, telemetry-estimated model-call cost and latency.

To achieve this aim, the research pursues the following objectives:

- 1) To critically review literature on next-generation long-context LLMs, long-context limitations, RAG-based prompting, retrieval failure modes, financial-document reasoning, and evidence-grounded inconsistency detection.
- 2) To construct an evidence-grounded evaluation design using a locked SEC 10-K filing set, a validated claim-case manifest, and the classification labels Supported, Contradicted, and Insufficient Evidence.
- 3) To implement comparable direct long-context and RAG-based context-delivery strategies using the selected GPT-5.5 API model while holding the filing corpus, SEC 10-K claim-cases, prompt structure, label definitions, output schema, scoring procedure, and telemetry-capture procedure constant.

- 4) To evaluate both context-delivery strategies empirically using multi-class classification metrics, structured qualitative error and evidence-use analysis, and programmatic logging of token use, telemetry-estimated model-call cost, and latency.
- 5) To synthesise the findings and assess whether any performance differences are justified by the corresponding operational trade-offs and practical value for SEC 10-K reasoning tasks.

1.4. Experimental Governing Terminology

This dissertation uses the following governing terms consistently:

- 1) *Controlled same-model comparison* refers to the overall experiment in which the selected GPT-5.5 API model is evaluated under both context-delivery strategies.
- 2) *Context-delivery strategy* refers to the independent variable, represented by the Long_Context and RAG-based strategies.
- 3) *Controlled experimental elements* refers to the fixed design components held constant across both strategies: the selected model, SEC 10-K filing corpus, claim-cases, task instructions, label definitions, output schema, scoring procedure and telemetry-capture procedure.

Under the *Long_Context* strategy, the model receives the full cleaned SEC 10-K filing. Under the *RAG-based* strategy, the same model receives locked retrieved chunks generated from the same filing corpus. These terms are used to avoid treating Long_Context and RAG-based as different models rather than as two context-delivery strategies applied to the selected model.

1.5. Research Questions

The dissertation is guided by three research questions:

RQ1: Does the Long_Context strategy improve cross-section inconsistency detection in SEC 10-K filings compared with the RAG-based strategy using the selected GPT-5.5 API model?

RQ2: What qualitative error and evidence-use patterns occur under the Long_Context and RAG-based strategies?

RQ3: Are any observed performance differences between the Long_Context and RAG-based strategies justified by the corresponding token use, telemetry-estimated model-call cost and latency trade-offs?

1.6. Scope and Research Design Overview

This dissertation is a *Type 5* experimental study with associated software development. The software serves as the reproducible experimental apparatus rather than an enterprise

financial-analysis application, supporting SEC 10-K acquisition and preparation, construction of the filing-specific overlapping chunk corpus and pre-generation of locked RAG contexts, strategy-specific prompt assembly, model execution, structured-output capture, scoring and telemetry logging. The study compares two context-delivery strategies using the selected GPT-5.5 API model:

- Under the Long_Context strategy, the model receives the full cleaned filing.
- Under the RAG-based strategy, the model receives retrieved chunks from the same filing corpus.

The selected model, filing corpus, claim-cases, task instructions, label definitions, output schema, scoring procedure and telemetry-capture procedure are held constant, while context delivery is varied. Outputs are classified as Supported, Contradicted or Insufficient Evidence.

Evaluation combines *quantitative* multi-class classification metrics, including accuracy, macro F1, and label-level precision, recall and F1-score (Sokolova and Lapalme, 2009), with *qualitative* analysis of error and evidence-use patterns using the procedure defined in Section 3.10 and the retained taxonomy and error evidence in Appendix E, Section E.7. Operational evaluation includes token use, telemetry-estimated model-call cost and latency. The experiment is bounded to six official SEC 10-K filings, 48 locked claim-cases, one selected GPT-5.5 API model and one locked RAG configuration.

1.7. Expected Contribution

The dissertation contributes a *controlled same-model comparison* that avoids the misleading “LLM versus RAG” framing; a reproducible SEC 10-K claim-case evaluation apparatus with formal scoring; and an integrated analysis of claim-classification performance, qualitative error and evidence-use patterns, token use, telemetry-estimated model-call cost and latency. It also distinguishes failures caused by incomplete context delivery from failures to apply available evidence under the locked scoring procedure.

These contributions are bounded to the selected GPT-5.5 API model, six-filing corpus, locked claim-case manifest, RAG configuration and experimental task. They do not establish general superiority for either context-delivery strategy across financial-document reasoning tasks.

The next chapter critically reviews prior work on long-context models and retrieval-augmented prompting to locate the dissertation’s synthesis gap and explain why a controlled same-model comparison for SEC 10-K cross-section inconsistency detection is required.

Chapter 2: Literature Review

2.1. Review Approach, Scope and Structure

The review adopts a structured narrative approach informed by PICOC principles without claiming full systematic-review status. It provides a controlled thematic synthesis of literature relevant to long-context models and their limitations, RAG-based prompting, retrieval limitations, financial-document reasoning, evidence grounding, inconsistency detection and evaluation methods.

Appendix A1 records the review protocol, including the PICOC framing, search themes, screening criteria, extraction fields, synthesis method and citation-validation rules. Appendix A2 records the validated source set, source roles, supporting pinpoints and citation boundaries used to control source use across dissertation report. Only sources recorded through this process were retained for citation.

The literature review is organised thematically rather than chronologically. The sequence moves from the *ROUS04* long-context motivation to document-scale reasoning and its limitations, then to RAG as the comparative baseline and its retrieval failure modes. It subsequently examines SEC 10-K financial-document reasoning, evidence grounding and inconsistency detection before synthesising prior evaluation methods and the remaining research gap.

Table 2.1: Summarises the PICOC scope governing the review.

PICOC element	Dissertation framing
Population	Official SEC 10-K filings, with related financial long-document studies as supporting context
Intervention	Direct long-context prompting using the full cleaned filing
Comparison	RAG-based prompting using retrieved chunks from the same cleaned filing corpus
Outcomes	Claim-classification performance, qualitative error and evidence-use patterns, token use, telemetry-estimated model-call cost and latency
Context	Financial long-document reasoning under a Type 5 experimental design

2.2. ROUS04 Theme and the Long-Context Motivation

This dissertation is grounded in the *ROUS04* project theme, which examines how next-generation long-context language models can be applied to real-world tasks and whether million-token context windows offer measurable benefits. The theme shifts attention from short-prompt demonstrations to document-scale reasoning, where lengthy books, codebases or regulatory documents may be supplied within a model's context window when technically feasible.

This motivation aligns with wider LLM research. General LLM surveys describe architectural developments enabling expanded context windows, and long-context benchmarks evaluate models on extended-sequence and document-level tasks (Naveed et al., 2025; Bai et al., 2024; An et al., 2024). However, processing extensive inputs can increase inference overhead, making token use, latency and operational cost relevant evaluation considerations for this experiment (Agrawal et al., 2024).

The dissertation operationalises the *ROUS04* theme through a same-model comparison of full-filing and retrieved-chunk context delivery. The comparison evaluates claim-classification performance, qualitative error and evidence-use patterns, token use, telemetry-estimated model-call cost and latency. The central issue is not whether full-filing prompting is technically possible, but whether it provides measurable benefit for SEC 10-K cross-section inconsistency detection when interpreted alongside its operational burden.

2.3. Long-Context Models and Document-Scale Reasoning

Long-context evaluation has developed because conventional short-context benchmarks do not adequately test whether models can process and use information distributed across extensive inputs. *LongBench* evaluates long-context understanding across document question answering, multi-document question answering, summarisation, few-shot learning and synthetic tasks, while *L-Eval* proposes standardised datasets and metrics for long-input evaluation (Bai et al., 2024; An et al., 2024).

These benchmarks establish the technical opportunity to supply full documents directly to an LLM rather than limiting the model to isolated excerpts. This is relevant to SEC 10-K filings, which are structured and cross-referential documents linking Business, Risk Factors, Management’s Discussion and Analysis, Market Risk, financial statements and notes (U.S. Securities and Exchange Commission, 2026). The Long_Context strategy may therefore expose relationships between evidence distributed across multiple filing sections without first restricting the model to a retrieved subset.

However, these benchmark findings cannot be transferred uncritically to SEC 10-K cross-section inconsistency detection. *LongBench* and *L-Eval* evaluate general long-context task families rather than the dense, cross-referential financial-disclosure structure, explicit three-label classification task and controlled same-model comparison used in this dissertation. These benchmarks therefore establish the feasibility and evaluation landscape for document-scale prompting; they do not demonstrate that the Long_Context strategy offers measurable advantages over the RAG-based strategy in this task.

2.4. Limitations of Long-Context Models

Large context windows do not guarantee effective evidence use across extended inputs. Long-context studies identify positional effects, including the “lost in the middle” effect, whereby relevant evidence may be underused despite being present in the supplied context (Liu et al., 2024; Hsieh et al., 2024). This is relevant to SEC 10-K reasoning because evidence required for claim assessment may be distributed across multiple filing sections (U.S. Securities and Exchange Commission, 2026).

Nominal token capacity must therefore be distinguished from effective context use. *RULER* evaluates how far models can use information within extended contexts rather than only how many tokens they can accept (Hsieh et al., 2024). *NeedleBench* similarly evaluates retrieval and reasoning over large-context inputs (Li, M. et al., 2024). However, their reliance on synthetic or benchmark-specific tasks limits their ability to reproduce the semantic and cross-referential complexity of official financial disclosures.

These limitations do not establish that long-context prompting is ineffective. They instead motivate direct evaluation of whether the selected model can distinguish evidence that supports a claim, directly contradicts it, or remains insufficient when relevant information is distributed across an SEC 10-K filing.

2.5. Retrieval-Based Prompting as a Comparative Baseline

RAG-based prompting provides an alternative, retrieval-mediated, input-length-reducing approach to long documents by supplying selected evidence rather than the complete source text (Lewis et al., 2020; Gao et al., 2023; Li, Z. et al., 2024). The original RAG formulation combines a parametric generator with non-parametric retrieved evidence, conditioning generation on retrieved passages rather than model weights alone (Lewis et al., 2020). Later work describes RAG more broadly as a retrieval, augmentation and generation workflow for knowledge-intensive tasks (Gao et al., 2023).

In this dissertation, RAG is not a competing model family but an alternative context-delivery strategy applied to the selected GPT-5.5 API model. The Long_Context strategy supplies the full cleaned filing, whereas the RAG-based strategy supplies retrieved chunks from the same filing corpus. The comparison therefore examines whether the same model behaves differently under full-filing and retrieved-chunk context delivery, rather than asking whether “LLMs are better than RAG.”

Recent comparative research treats long-context prompting and RAG as alternative mechanisms whose relative value can depend on task complexity, context length, retrieval quality and operational burden. Li, Z. et al. (2024) compare these approaches directly,

supporting task-specific empirical evaluation rather than assuming either strategy is universally preferable.

2.6. RAG Limitations: Retrieval Miss, Noise and Fragmentation

RAG-based prompting can reduce input length and focus the model on selected evidence, but it also makes performance dependent on retrieval quality. If retrieval misses the relevant section, selects an incorrect passage or returns noisy near-matches, the model receives an incomplete or distorted evidence base. Chunking may also fragment key sentences or separate financial values from their headings, notes or explanatory context.

These concerns are supported by retrieval research. Cuconasu et al. (2024) show that passage relevance, quantity and position can affect generation quality, including degradation caused by noisy or hard-negative passages. Barnett et al. (2024) identify failure points across retrieval, augmentation and generation, while RAG surveys emphasise evaluating retrieval and generation quality rather than relying on fluent final answers alone (Yu et al., 2024; Gao et al., 2023).

These limitations are particularly relevant to SEC 10-K cross-section reasoning, where claim assessment may require evidence from multiple filing sections. Retrieving only part of the required evidence can produce an unsupported classification even when the complete filing contains sufficient information. Retrieval miss, noise and fragmentation are therefore treated as qualitative error considerations in the experiment.

2.7. Financial Document Reasoning and SEC 10-K Filings

SEC 10-K filings form the document population for this experiment. They are official, public and systematically structured annual disclosures covering business operations, risk factors, financial condition and audited financial statements (U.S. Securities and Exchange Commission, n.d.; U.S. Securities and Exchange Commission, 2026). SEC investor guidance also emphasises critical reading of disclosures whose explanation or clarity may be deficient or inconsistent with disclosure requirements (U.S. SEC Office of Investor Education and Advocacy, 2011). These characteristics support evaluating evidence integration across filing sections rather than isolated-passage reasoning.

Financial NLP research further illustrates how specialised forms of evidence use are evaluated in financial documents. *FinanceBench* evaluates evidence-grounded question answering over public-company documents (Islam et al., 2023), while *SEC-QA* examines question answering over SEC filings and reports retrieval-related limitations in realistic financial settings (Lai et al., 2025). *DocFinQA* extends financial question answering to full-document context (Reddy et al.,

2024), and *TAT-QA* and *FinQA* demonstrate the need to combine textual, tabular and numerical evidence (Zhu et al., 2021; Chen et al., 2021).

These benchmarks are relevant but not identical to the present task. They primarily evaluate question answering, fact retrieval or numerical reasoning, whereas this dissertation examines evidence-grounded cross-section inconsistency detection under the Long_Context and RAG-based strategies in a controlled same-model comparison. Numeric-consistency claim-cases are included within that broader claim-classification task rather than treated as standalone numerical question answering.

SEC 10-K filings are therefore suitable because they are lengthy, real-world regulatory disclosures with a cross-referential structure that distributes potentially relevant evidence across Business, Risk Factors, Management’s Discussion and Analysis, Market Risk, financial statements and notes. This provides an appropriate setting for comparing the Long_Context and RAG-based strategies for evidence-grounded cross-section inconsistency detection.

2.8. Evidence Grounding, Hallucination and Inconsistency Detection

Evidence-grounded financial inconsistency detection requires model classifications and rationales to remain tied to the supplied document evidence. Hallucination research identifies the possibility that LLM outputs may be unsupported by or inconsistent with available evidence (Huang et al., 2023). In this dissertation, that literature motivates assessment of whether classifications and rationales are justified by the SEC 10-K evidence supplied under each context-delivery strategy; hallucination is not treated as a separate experimental outcome.

Financial claim-verification research provides a closer methodological connection. *FinDVer* treats claims, evidence and verification outcomes as explicit evaluative objects within long, hybrid-content financial documents (Zhao et al., 2024), while financial contradiction-detection research examines semantic conflict in financial reports (Deußner et al., 2023). These studies support a claim-evidence-decision framing for SEC 10-K inconsistency detection.

This framing motivates the dissertation’s three-way distinction between Supported, Contradicted and Insufficient Evidence. Supported requires filing evidence that justifies the claim; Contradicted requires direct conflict; and Insufficient Evidence applies where the available context is incomplete, ambiguous, overly general or non-specific. This distinction is essential because absence of sufficient evidence does not establish contradiction.

2.9. Prior Research Methods in Related Work: A Synthesis

Prior studies use multiple complementary methodological approaches. Long-context research commonly employs benchmark datasets, synthetic retrieval tasks, long-document question

answering and stress tests to examine variation by input length, task type and evidence position (Bai et al., 2024; An et al., 2024; Liu et al., 2024; Hsieh et al., 2024). RAG evaluation increasingly separates retrieval relevance, context quality, answer faithfulness and answer relevance to identify failures that final-answer fluency alone may conceal (Es et al., 2024; Yu et al., 2024; Saad-Falcon et al., 2024).

Within financial NLP, studies typically use benchmark datasets, task-specific labels and explicit evidence references for question answering, numerical reasoning and claim verification (Islam et al., 2023; Lai et al., 2025; Reddy et al., 2024; Zhao et al., 2024). This methodological precedent supports a claim-case classification design with discrete labels, preserved filing-reference anchors and qualitative error and evidence-use analysis.

These task designs also require evaluation beyond raw accuracy. Precision, recall and F1-score are important because accuracy can obscure label-specific weaknesses (Sokolova and Lapalme, 2009). In a multi-class classification task, accuracy alone may conceal whether a system correctly identifies Contradicted claim-cases while mishandling Supported or Insufficient Evidence boundaries. Class-specific metrics, confusion matrices and false-positive/false-negative analysis are therefore appropriate. Moreover, broader LLM-evaluation and inference-systems literature also supports measuring efficiency, cost and latency alongside task scores (Liang et al., 2022; Agrawal et al., 2024). Accordingly, token use, telemetry-estimated model-call cost and latency are logged as dependent measures to assess the practical trade-offs of large-context prompting.

Finally, these evaluation choices must be implemented within a reproducible experimental design. Experimental software-engineering literature emphasises clearly defined designs, variables and validity controls (Wohlin et al., 2012). LLM-evaluation studies also warn that benchmark results can be affected by stochastic behaviour, repeated-run uncertainty and evaluation conditions (Blackwell et al., 2024). These concerns support a design that holds the selected model, filing corpus, claim-cases, task instructions, label definitions, output schema, scoring procedure and telemetry-capture procedure constant while varying only the context-delivery strategy.

2.10. Research Gap and Chapter Summary

The reviewed literature establishes a coherent but incomplete foundation for this dissertation. Long-context models make full-document prompting technically feasible. However, evidence-position effects on models' use of relevant information, together with the distinction between nominal context capacity and effective evidence-use, show that expanded capacity does not guarantee accurate use of evidence distributed across a document. RAG-based prompting

offers a retrieval-mediated, input-length-reducing alternative, but its performance depends on retrieval relevance, completeness and resistance to noise or chunk fragmentation.

SEC 10-K filings provide an appropriate setting for examining these limitations because they are lengthy, structured, cross-referential regulatory disclosures in which claim assessment may require evidence from multiple sections. Existing financial-NLP benchmarks evaluate combinations of textual, tabular, numerical and evidence-grounded reasoning, but predominantly focus on question answering, fact retrieval, numerical reasoning or related claim-verification tasks rather than controlled cross-section inconsistency detection. Claim assessment also requires an explicit distinction between evidence that supports a claim, evidence that directly contradicts it and context that is insufficient to justify either conclusion.

The *resulting research gap* is therefore specific: existing literature does not determine whether direct long-context prompting offers measurable advantages over RAG-based prompting for SEC 10-K cross-section inconsistency detection when the same model, filing corpus, claim-cases, task instructions, label definitions, output schema and scoring procedure are held constant. Nor does it establish whether any claim-classification performance difference justifies the associated token use, telemetry-estimated model-call cost and latency trade-offs. This dissertation addresses that gap through a reproducible Type 5 experiment comparing two context-delivery strategies using the selected GPT-5.5 API model. Context-delivery strategy is the independent variable, while claim-classification performance, qualitative error and evidence-use patterns, token use, telemetry-estimated model-call cost and latency form the dependent measures. The fixed controls comprise the selected model, filing corpus, claim-case manifest, task instructions, label definitions, output schema, scoring procedure, retry rules and telemetry-capture procedure.

Table 2.2: Methodological Mapping from Literature to Experimental Design

Literature area	What it establishes	Key limitation	Design implication
Long-context evaluation	Extended inputs enable document-scale prompting.	Capacity does not guarantee effective evidence use.	Include Long_Context strategy as full-filing context delivery.
Long-context limitations	Evidence position and task complexity can affect performance.	Findings are often synthetic or not SEC 10-K specific	Analyse long-context evidence-use failures qualitatively.
RAG and retrieval evaluation	Retrieval can focus and reduce the context supplied to the model.	Retrieval may miss, fragment or distort relevant evidence.	Use RAG-based strategy as the comparison strategy and track retrieval-related errors.
Financial-document reasoning	SEC 10-K filings are structured, lengthy, and evidence-rich.	Prior benchmarks mainly evaluate related question-	Use SEC 10-K filings and explicit cross-section claim-cases.

Literature area	What it establishes	Key limitation	Design implication
		answering and numerical tasks.	
Evaluation methods	Classification quality, evidence use and operational efficiency require multiple measures.	Accuracy alone can conceal class-specific weaknesses.	Use class-specific metrics, qualitative error and evidence-use analysis, token use, cost, and latency.

The next chapter defines the research design, controlled experimental elements, dataset, claim-case construction, software apparatus, execution procedure and evaluation methods.

Chapter 3: Methodology

This dissertation adopts a *Type 5* experimental design with associated software development to evaluate two context-delivery strategies - Long_Context and RAG-based - within a *controlled same-model comparison*. *Context-delivery strategy* is the independent variable, while the *controlled experimental elements* remain fixed across both strategies.

The methodology follows a controlled sequence comprising filing-corpus preparation, claim-case construction, pre-generation and locking of RAG contexts, strategy-specific prompt assembly, same-model execution, structured-output and telemetry capture, post-execution scoring, and qualitative error and evidence-use analysis.

The associated software serves as the reproducible experimental apparatus rather than as an enterprise financial-analysis product. It supports controlled execution and preserves an auditable path from locked inputs to retained outputs.

This chapter defines the research design, methodological controls and evaluation procedure. Chapter 4 documents the implementation and execution of the apparatus. Pilot calibration outputs were used for method validation only and were not included as final dissertation findings.

3.1. Research Design

The study uses a controlled same-model experimental design combining *quantitative* performance evaluation with *qualitative* error and evidence-use analysis. The quantitative component compares claim-classification performance, token use, telemetry-estimated model-call cost and latency across the Long_Context and RAG-based strategies. The qualitative component examines whether claim-classification decisions and supporting rationales remain grounded in SEC 10-K evidence and categorises incorrect outputs using the predefined qualitative taxonomy and diagnostic evidence retained in Appendix E, Section E.7, Tables E.12a-E.16. Claim-case type context is retained in Appendix C, Table C.5a.

The design is driven by the three research questions rather than intended as a general benchmark of model capability.

The experimental objects, context-delivery strategies, controlled experimental elements and evaluation measures were defined before the main run to support a fair comparison and mitigate validity threats, consistent with experimental software-engineering principles (Wohlin et al., 2012). Table 3.1 maps each research question to its corresponding methodological response and evidence source.

Table 3.1: Methodological Mapping from Research Questions to Experimental Design

Research question	Methodological response	Primary evidence produced
RQ1: Claim-classification performance	Execute each of the 48 locked SEC 10-K claim-cases once under the Long_Context and once under the RAG-based strategy, using the selected GPT-5.5 API model and controlled experimental elements.	Accuracy and macro F1; label-level precision, recall and F1-score; confusion matrices; case-level correctness and strategy differences.
RQ2: Qualitative error and evidence-use patterns	Review incorrect or insufficiently supported outputs using the predefined qualitative error taxonomy and compare evidence-reference use, evidence grounding, error direction, and shared or strategy-specific error patterns.	Error categories and frequencies; error-direction distributions; evidence-grounding observations; shared and strategy-specific case-level diagnoses.
RQ3: Operational trade-offs	Record input, output and total token use, API latency and telemetry-estimated model-call cost for every strategy-level call.	Per-call and strategy-level token, cost and latency summaries interpreted alongside claim-classification performance.

3.2. Experimental Variables, Outcomes and Controlled Elements

Context-delivery strategy is the sole independent variable, represented by the Long_Context and the RAG-based strategies. The experiment evaluates three groups of dependent measures.

1. First, quantitative claim-classification performance is assessed using accuracy and macro F1, together with label-level precision, recall and F1-score, confusion-matrix counts and case-level correctness.
2. Second, structured qualitative analysis examines error direction, predefined qualitative error categories, evidence-reference use, evidence-grounding observations and shared or strategy-specific error patterns.
3. Third, operational evaluation measures input, output and total token use, telemetry-estimated model-call cost and API latency.

The controlled experimental elements in Table 3.2 are held constant across both strategies. Holding these elements constant reduces the likelihood that outcome differences arise from changes in the model, filing corpus, task definition or scoring procedure rather than from context-delivery strategy.

Table 3.2: Independent Variable and Controlled Experimental Elements

Experimental component	Long_Context strategy	RAG-based strategy
Selected model	selected GPT-5.5 API model	Same as Long_Context
Source filing corpus	Same six cleaned SEC 10-K filings	Same as Long_Context

Experimental component	Long_Context strategy	RAG-based strategy
Context supplied	Full cleaned filing	Eight locked retrieved chunks generated from the same cleaned filing corpus
Controlled experimental elements	Same claim-cases, task instructions, label definitions, output schema, scoring procedure, telemetry-capture procedure and provider-exposed request settings	Same as Long_Context

The temperature parameter was omitted from GPT-5.5 API requests for both strategies because the final Responses API path rejected it.

3.3. Dataset Selection

The dataset comprises six official SEC Form 10-K filings for Apple, Amazon, JPMorgan Chase, Bank of America, Verizon and Ford. These structured annual disclosures distribute relevant evidence across business descriptions, risk factors, Management’s Discussion and Analysis, market-risk disclosures, financial statements and notes. Each filing was retained only after passing acquisition, extraction, section-coverage, token-feasibility and RAG-chunking validation checks.

The filings were purposively selected to provide variation in disclosure structure and document complexity while retaining a bounded, manually auditable corpus appropriate to the dissertation scope. The corpus covers technology, retail, cloud, financial-sector, telecommunications, regulatory, automotive, manufacturing and financing-arm disclosure patterns. The corpus was not intended to be statistically or sector-representative.

This selection supports construction of cross-section claim-cases across varied filing lengths, business models and filing sections. Estimated full-filing inputs ranged from approximately 55,263 tokens for Apple to 353,892 tokens for JPMorgan Chase. Table 3.3 summarises cleaned filing sizes and planning-stage token estimates. Appendix C, Section C.2 retains the complete filing register, accession metadata, source URLs, detailed measurements, chunk counts and smoke-test status.

Because the filings were extracted from EDGAR HTML/iXBRL pages rather than page-native PDFs, rendered page count was not used as a feasibility criterion. Feasibility was assessed using cleaned text, estimated input-token boundaries and section-coverage checks before main-run execution.

Table 3.3: Main-Run Filing Size and Estimated Input Tokens

SEC 10-K filing	Fiscal year end	Character / Word count	Input tokens
Apple Inc.	2025-09-27	221,051 / 32,213	~55,263

SEC 10-K filing	Fiscal year end	Character / Word count	Input tokens
Amazon.com, Inc.	2025-12-31	313,252 / 44,517	~78,313
JPMorgan Chase & Co.	2025-12-31	1,415,566 / 190,576	~353,892
Bank of America Corp.	2025-12-31	1,175,470 / 147,362	~293,868
Verizon Communications Inc.	2025-12-31	536,983 / 72,885	~134,246
Ford Motor Co.	2025-12-31	753,575 / 104,185	~188,394

3.4. Claim-Case Construction and Ground-Truth Definition

The experiment evaluates model behaviour through structured claim-cases. Each case contains a natural-language claim concerning one SEC 10-K filing, associated filing metadata, an expected label, a ground-truth rationale and filing-evidence anchors. The same claim text and metadata were used under both context-delivery strategies.

The locked manifest comprises 48 claim-cases - eight per filing - with a balanced distribution of 16 Supported, 16 Contradicted and 16 Insufficient Evidence cases. All claim-cases, expected labels, rationales and evidence anchors were constructed, reviewed and locked before the main run, preventing post-execution adjustment of claims or scoring labels.

Formal label definitions are given in Table 3.4. Absence of support was not treated as contradiction. This three-label structure supports evidence-grounded inconsistency detection while limiting unsupported inference beyond the filing evidence.

Table 3.4: Formal Label Definitions for Claim Classification

Label	Definition	Primary classification risk
Supported	Evidence sufficiently and directly justifies the claim.	Accepting related, adjacent or partial evidence as sufficient support.
Contradicted	Evidence directly conflicts with the claim semantically, numerically, temporally or at segment level.	Treating absence of support as contradiction.
Insufficient Evidence	Evidence is absent, incomplete, ambiguous, overly general or insufficiently specific to establish support or contradiction.	Overusing the label or making unsupported inference

Expected labels, ground-truth rationales and filing-evidence anchors were retained in the hidden scoring register and excluded from model-facing prompts. Appendix C contains the locked claim-case manifest, expected labels, rationales and evidence anchors.

In addition to label balance, the manifest deliberately spread claim-cases across specialised case-type groups to test different evidence-use demands, including cross-section synthesis,

numerical consistency, segment consistency, risk-to-MD&A linkage and evidential insufficiency. The machine-readable `cross_section_synthesis` type represents one general synthesis category rather than the full scope of cross-section inconsistency detection. Appendix C, Table C.5a records the locked distribution of claim-cases across specialised case-type groups.

3.5. Long_Context Strategy: Full-Filing Prompting

Under the Long_Context strategy, the selected GPT-5.5 API model receives the full cleaned text of the relevant SEC 10-K filing. Pre-main-run feasibility and smoke testing confirmed that all six filings could be supplied without reducing them to selected sections; their estimated token lengths are reported in Table 3.3. This strategy evaluates whether full-filing context enables the model to identify and integrate evidence distributed across filing sections without retrieval-stage filtering.

The model is instructed to classify each claim using only the supplied filing evidence and not external knowledge. No manually selected evidence excerpts are added to the Long_Context input because this would introduce retrieval-like preselection and alter the context-delivery strategy being evaluated.

Appendix D retains the Long_Context prompt template and output schema, while Appendix C records the filing-feasibility evidence.

3.6. RAG-Based Strategy: Retrieval-Mediated Prompting

Under the RAG-based strategy, the selected model receives pre-generated and locked retrieved chunks from the same cleaned SEC 10-K corpus used by Long_Context. Retrieval queries were derived solely from the locked natural-language claim. Expected labels, ground-truth rationales, evidence anchors and other scoring-key information were excluded from retrieval and model-facing inputs to prevent answer-key leakage.

During preparation, the locked RAG configuration used text-embedding-3-large, fixed 750-word windows, 100-word overlap, in-memory cosine-similarity ranking and top-k retrieval of eight chunks per claim-case. The six filings produced 913 chunks. Appendix D, Section D.6 records the complete RAG configuration, while Appendix C, Section C.2 records the filing-level chunk distribution.

Each locked context retained identifiers, ranks, similarity scores, word ranges and supplied text. The pre-generated contexts are retained in `inputs/rag/rag_retrieved_contexts_locked_v1.jsonl`, with Appendix E, Sections E.2 and E.4 linking the locked inputs to the executed RAG-based calls.

Pre-main-run retrieval calibration assessed whether the locked configuration returned relevant evidence before scored execution. After the contexts were locked, they were not modified in response to model outputs. Retrieval miss, noise, incomplete evidence and chunk-boundary fragmentation therefore remained possible properties of the RAG-based strategy rather than being corrected post hoc.

3.7. Prompt Template and Structured Output Schema

Both context-delivery strategies used versioned prompt templates with the same task instructions, label definitions and strict JSON output schema. The supplied context was the only strategy-specific difference: the full cleaned filing for the Long_Context strategy and the locked retrieved chunks for the RAG-based strategy. The model was instructed to classify each claim using only the supplied SEC 10-K evidence, avoid external knowledge and return valid JSON. Prompt versions were locked after pilot calibration and before the main run, preventing changes in response to main-run outputs.

The shared output schema required `case_id`, `predicted_label`, `evidence_references`, `reasoning_summary` and `confidence`. `predicted_label` supplied the value used for claim-classification scoring. `evidence_references` and `reasoning_summary` supported post-execution examination of evidence use and grounding. `confidence` was retained as a descriptive output field only and was not used for claim-classification scoring.

The GPT-5.5 Responses API enforced the strict JSON schema, and outputs were parsed programmatically using predefined failure-handling rules. No separate schema-validity field was retained because all main-run outputs passed schema parsing. The subsequent qualitative analysis procedure is defined in Section 3.10.

Appendix D, Sections D.3-D.5 retain the executed prompt templates and shared schema. Appendix E, Section E.4 indexes the corresponding request payloads and retained raw and parsed outputs.

3.8. Preliminary Validation and Method Calibration

Before the main run, staged pilot and stress checks assessed filing preparation, full-filing feasibility, RAG retrieval, GPT-5.5 API execution, structured output capture and telemetry logging. These activities calibrated the experimental method and apparatus rather than producing final dissertation findings.

The pilot refined procedural boundaries without changing the research questions, locked claim-case design or scoring framework. It clarified the Insufficient Evidence boundary for unsupported causal claims and retrieval-completeness requirements before the main-run

lock. After the corpus, claim-cases, scoring key, prompts, RAG contexts and execution queue were locked, no adjustment based on main-run outputs was permitted.

Appendix F retains the pilot, stress-test, calibration and pre-main-run validation records as audit evidence. Main-run outputs and scoring evidence are retained separately in Appendix E.

3.9. Main Experiment Execution Procedure

Before the main run, the filing corpus, 48-claim-case manifest, hidden scoring key, model and request configuration, RAG settings, prompt and schema versions, retry rules, logging procedure and cost controls were formally locked. Each claim-case was executed once under the Long_Context strategy and once under the RAG-based strategy without outcome-driven modification.

The main run produced 96 strategy-level model calls: 48 claim-cases across two context-delivery strategies. One predefined retry was permitted only for technical failure using the unchanged prompt and supplied evidence. Claim-classification errors, weak rationales and post-output retrieval misses were not retried or corrected.

Table 3.5: Main-Experiment Computational and Analytical Sequence

Step	Main-run computational and analytical action
1	Confirm the locked filing corpus, claim-case manifest, hidden scoring key, model and request settings, RAG configuration, prompt and schema versions, retry rules, logging procedure and cost controls.
2	Assemble Long_Context requests using the full cleaned filing and load the pre-generated, locked RAG context for each corresponding claim-case.
3	Execute each claim-case once under both context-delivery strategies through the controlled API runner, applying only the predefined technical-failure retry rule.
4	Retain request payloads, hashes, raw responses, structured outputs, run identifiers and telemetry, while keeping the hidden scoring key separate from model-facing execution.
5	Introduce the hidden scoring key after execution, score predicted labels by exact match, and derive classification metrics, token use, telemetry-estimated model-call cost and latency.
6	Apply the predefined post-scoring qualitative taxonomy and interpret the quantitative, qualitative and operational evidence against RQ1-RQ3 within the locked experimental boundaries.

Appendix B retains the complete parameter-lock, retry and execution protocol. Appendices C-F retain the locked inputs, prompt and RAG configuration, main-run evidence, and pilot and reproducibility records.

3.10. Evaluation Metrics and Analysis Procedure

RQ1 was evaluated by comparing each `predicted_label` with the locked expected label for the corresponding claim-case. Quantitative measures comprised overall accuracy and macro F1, together with label-level precision, recall and F1-score, confusion matrices and case-level correctness. Accuracy summarised overall performance, while macro F1 and label-level measures showed whether performance differed across Supported, Contradicted and Insufficient Evidence (Sokolova and Lapalme, 2009).

RQ2 was evaluated after quantitative scoring through structured review of incorrect outputs and evidence-use issues. Each case was examined using its expected and predicted labels, claim-case type, supplied evidence references and reasoning summary; for RAG-based errors, the review also considered whether retrieval miss, incomplete evidence, noise or chunk-boundary fragmentation contributed to the outcome. The researcher coded errors by classification direction and evidential cause using the predefined qualitative taxonomy and case-level diagnostic evidence retained in Appendix E, Section E.7, Tables E.12a-E.16. Claim-case type context was taken from the locked manifest distribution in Appendix C, Table C.5a. This post-scoring diagnosis described observed error patterns but did not alter the original prediction or quantitative score. The analysis distinguished claim-case type from qualitative error family. Claim-case type was assigned before execution in the locked claim-case manifest to describe the designed evidence-use demand of each case. Qualitative error family was assigned only after scoring to describe the observed failure in a misclassified output. These fields were not treated as equivalent or as a fixed mapping; they were joined at row level to examine whether particular designed claim-case types concentrated particular post-scoring error families.

RQ3 was evaluated using input, output and total token use, telemetry-estimated model-call cost and API latency. These measures were aggregated by context-delivery strategy and interpreted alongside claim-classification performance and qualitative error patterns.

The three analytical components were interpreted jointly, and conclusions were bounded to the selected model, six filings, 48 locked claim-cases, prompt design and RAG configuration. Appendix C retains the label-decision and scoring rules, while Appendix E retains the final scoring records, metrics, confusion matrices, qualitative coding and telemetry evidence.

3.11. Validity, Reliability, Replicability and Generalisability

Internal validity was supported through a controlled same-model comparison: both strategies used the same selected GPT-5.5 API model identifier, locked request configuration and provider-exposed settings, with context-delivery strategy as the sole independent variable. Remaining threats included researcher judgement in claim-case construction, dependence on

the locked RAG configuration, pilot-calibration effects and unobservable provider-side model changes. Mitigations included formal parameter locks, paired execution, predefined label rules, documented rationales and evidence anchors, exclusion of ambiguous cases, separation of pilot and main-run evidence, and retention of provider identifiers and timestamps.

Construct validity was strengthened by designing the task around evidence-grounded cross-section inconsistency detection rather than term matching. Claim-cases included distributed-evidence, numerical-dependency, Insufficient Evidence and deliberately contradicted cases. Because genuine contradictions in audited filings are rare, Contradicted cases represented conflicts with filing evidence rather than allegations of misreporting. Label definitions, rationales and anchors were fixed pre-execution, while expected labels and scoring information were withheld from model inputs.

Reliability and *replicability* were supported through consistent procedures, programmatic parsing, exact label scoring and retention of experimental artefacts: filing/extraction records, locked claim-case manifest, prompt templates, output schema, RAG configuration, retrieved-context records, execution manifests, payloads, raw/parsed responses, telemetry and scoring outputs. These provide an auditable chain from each claim-case to its prediction. However, each case was executed once per strategy and the provider-hosted API did not permit claims of deterministic repeatability; the study therefore demonstrates procedural reliability rather than run-to-run determinism.

Conclusion validity and *generalisability* remain bounded. The quantitative comparison is descriptive rather than inferential; no statistical significance or universal superiority is claimed. Findings are limited by the purposive selection of six filings, one selected model, 48 claim-cases, one prompt design and one RAG configuration, though observed patterns may inform broader studies. Claim-case construction and qualitative coding were researcher-derived without independent dual coding, so some judgemental bias may persist.

Appendices B-F retain the execution protocol, locked inputs, prompt and RAG configuration, main-run evidence, pilot records and audit materials.

3.12. Ethical and Data-Handling Considerations

The experimental corpus comprised publicly available SEC Form 10-K filings. The study did not intentionally collect private personal data, confidential corporate information, proprietary analyst research or human-participant data. Names and officer details contained in the filings were treated solely as elements of public regulatory disclosures and were not analysed as personal data or as attributes of human participants.

The Supported, Contradicted and Insufficient Evidence labels, model predictions and accompanying rationales were produced for a controlled computer-science experiment. They were not interpreted as investment advice, audit conclusions, legal determinations, financial recommendations or judgements concerning corporate integrity or misreporting. Deliberately contradicted claim-cases represented test statements that conflicted with filing evidence rather than allegations that the underlying filings were inaccurate.

API credentials and local secrets were managed through environment configuration and excluded from version control and submitted artefacts. The execution pipeline processed public filing text, claim-case prompts, supplied contexts and structured model outputs. Raw and parsed responses, telemetry, scoring records and audit evidence were retained to support auditability and procedural replicability. Appendix F retains the supporting ethics, data-handling and audit records.

The next chapter documents the implementation and execution of the locked experimental pipeline.

Chapter 4: Experimental Implementation and Execution

4.1. Experimental System and Locked-Corpus Overview

This chapter describes the implemented *Type 5* experimental apparatus and its execution rather than revisiting methodological rationale. The containerised command-line system operationalised the controlled same-model comparison between the Long_Context and RAG-based strategies through frozen inputs, validation gates and an auditable evidence-export workflow. The full codebase is submitted as 04_Codebase/CS958-R0US04-Codebase.zip, with all .py files under /app. Appendices B-F retain the execution protocol, locked inputs, prompt and RAG configuration, main-run outputs, reproducibility artefacts and audit materials.

The six SEC Form 10-K filings used in the main run were the locked cleaned corpus defined in Section 3.3 and Appendix C, Section C.2. Both strategies used this same corpus state during execution, ensuring no variation in filing source, version or extraction state between the Long_Context and RAG-based strategies.

4.2. Text Extraction and Section Handling

The *EDGAR HTML/iXBRL* filings were converted into locked cleaned text before the main run. In the submitted codebase, `filing_text_cleaner.py` implemented text preparation, while `section_inventory_builder.py` generated the filing-section inventory used to check coverage of Business, Risk Factors, Management's Discussion and Analysis, Market Risk, financial statements and notes.

`corpus_lock_validator.py` confirmed that every clean-text file declared by the locked execution queue existed before downstream processing. Appendix C, Section C.2 retains the extraction measurements and section-coverage records.

4.3. Full-Filing Feasibility and Input Validation

Before execution, `token_estimator.py` produced character-based feasibility estimates for each full cleaned filing used in the Long_Context strategy; these were treated as approximate indicators rather than tokenizer-exact counts. Appendix C, Section C.2 retains the filing measurements and estimated token lengths.

Prior to prompt generation, `input_artifact_validator.py` checked the consistency of all locked artefacts, including the 96 paired strategy-level rows, 48 scoring-key cases, balanced 16/16/16 label distribution, six cleaned filings, 48 frozen RAG contexts, matching queue and scoring-key identifiers, and hashes for the principal locked artefacts. `answer_key_leakage_checker.py` separately confirmed that expected labels, rationales, anchors and other scoring-key fields were excluded from the execution queue and model-

facing payloads. These validation gates ensured complete paired inputs and strict hidden-key separation. Appendix E, Section E.3 retains the locked-input and completion-validation evidence; Appendix F retains the readiness and leakage-control records.

4.4. RAG Corpus, Context Construction and Frozen-Context Boundary

The RAG-based preparation pipeline used the same six locked cleaned filings as the Long_Context strategy. `filing_chunker.py` split each filing into fixed 750-word windows with 100-word overlap, producing 913 chunks consistent with Appendix C, Section C.2. `openai_embedding_client.py` generated text-embedding-3-large embeddings for all chunks and locked claim texts. Chunk ranking used exact in-memory cosine similarity through `InMemoryCosineVectorStore` in `faiss_vector_store.py`; despite the filename, no FAISS index or external vector store was used. `rag_context_builder.py` retained the top eight chunks per claim-case, including identifiers, word ranges, ranks, similarity scores and supplied text.

These contexts were written as locked JSONL inputs before the main run. Retrieval smoke tests and manual evidence review checked for retrieval-configuration defects prior to lock. Although `rag_context_builder.py` supported regeneration when explicitly enabled, scored execution used `rag_context_loader.py` to load each pre-generated context by `run_id`, with no regeneration, manual repair or outcome-driven modification. Appendix D, Section D.6 retains the locked configuration and context-record structure; Appendix F retains the pre-lock calibration evidence.

4.5. Prompt Assembly and Staged Execution

The controlled workflow was orchestrated through `app/cli.py`. The `validate` command checked locked artefacts and hidden-key separation, while `prompt-dry-run` assembled all 96 model-facing payloads without issuing GPT-5.5 API calls. `LongContextPromptAssembler` inserted the full cleaned filing, and `RagPromptAssembler` inserted the frozen RAG context for each `run_id`; both used shared task instructions and label definitions from `PromptTemplateRegistry`. `PromptPayloadWriter` retained payloads, SHA-256 hashes and prompt-size summaries.

`smoke-live` executed one selected queue row, whereas `full-live` required the `--confirm-main-run` flag and was additionally gated by the default-disabled `ENABLE_LIVE_MODEL_CALLS` setting. Expected labels and rationales remained outside the execution queue and request payloads. Appendix D retains the locked prompt templates, output schema and RAG configuration; Appendix E retains validation, prompt-generation and execution evidence.

4.6. Structured Output Capture

Both context-delivery strategies used the same strict structured-output schema. `output_schema_registry.py` required `case_id`, `predicted_label`, `evidence_references`, `reasoning_summary` and `confidence`, restricted `predicted_label` to Supported, Contradicted or Insufficient Evidence, and rejected additional properties. The same schema was submitted through the Responses API for every model call.

For each completed call, `openai_responses_model_client.py` retained the complete provider response and extracted the returned JSON. `experiment_run_controller.py` wrote separate raw and parsed output records linked to the corresponding run, case and strategy identifiers. Strict schema enforcement and predefined parse-failure handling prevented invalid output from being silently accepted or manually repaired.

`model_output_parser.py` extracted predictions, confidence, evidence references and reasoning for scoring and qualitative review without altering the retained model output. Appendix D, Section D.5 defines the locked output schema, while Appendix E, Section E.4 indexes the 96 request payloads and corresponding raw and parsed outputs.

4.7. Scoring and Telemetry Logging

After full-live execution, `classification_scorer.py` loaded the retained parsed outputs and introduced the hidden scoring key through `scoring_key_repository.py`. Predictions were matched to expected labels by `case_id` and scored by exact label match without rewriting the model classifications or reasoning.

`metric_calculator.py` produced case-level scored records, strategy-level accuracy and confusion-matrix counts. Label-level precision, recall and F1-score, together with macro F1, were subsequently derived from the locked scoring outputs in the post-run analysis workbook.

For each execution row, `telemetry_logger.py` recorded execution status, elapsed time, input and output tokens, retries, error information and telemetry-estimated model-call cost. `cost_guard.py` calculated the cost estimate from token usage and configured input and output pricing constants, and enforced the cumulative hard-cost ceiling.

Appendix E, Section E.5 retains the per-call telemetry and aggregate token, cost and latency records. Appendix E, Section E.6 retains the scored predictions, strategy and label metrics, and confusion matrices. Reported costs are telemetry-estimated model-call costs rather than provider-billed per-call charges.

4.8. Reproducibility, Versioning and Implementation Boundaries

Reproducibility was supported through a containerised command-line apparatus, environment-driven configuration, frozen inputs and staged execution gates. The Dockerfile and Docker Compose configuration defined the container runtime and mounted the locked input and output paths. ExperimentRuntimeSettings centralised model and embedding identifiers, artefact paths, RAG settings, retry limits, cost controls and feature toggles. Live execution and optional preparation or RAG-context rebuilding remained disabled unless explicitly enabled.

RunManifestExporter recorded the active provider and model identifiers, locked artefact paths and feature toggles. Prompt hashes, raw and parsed outputs, telemetry and scoring records remained traceable to the locked inputs, while AppendixEExporter and EvidencePackBuilder assembled the principal evidence files and portable archive.

The provider-hosted GPT-5.5 API remained an implementation boundary because hidden provider-side changes could not be controlled. Temperature was omitted because the final Responses API path rejected the parameter. The same request path was used for both context-delivery strategies, and exact output determinism is not claimed. Appendix E retains the main-run evidence chain, while Appendix F retains the reproducibility and final-audit records.

The next chapter reports and interprets the claim-classification, qualitative error and operational telemetry findings produced by the main run.

Chapter 5: Analysis, Findings and Discussion

5.1. Evaluation Basis and Procedure

This chapter reports and interprets the main-run findings for six SEC Form 10-K filings, 48 locked claim-cases and 96 completed strategy-level calls: 48 under the Long_Context strategy and 48 under the RAG-based strategy. All calls completed with zero retries; no output was excluded, corrected or rerun.

The analysis uses the scoring and qualitative-review procedure defined in Chapter 3 and Appendix C. Results are aggregated by context-delivery strategy for claim-classification performance, qualitative error and evidence-use patterns, and operational telemetry. Appendix C retains the locked labels, scoring rules, claim-cases and claim-case type distribution, including Table C.5a. Appendix E retains the outputs, telemetry, metrics and qualitative error evidence, including the case-level diagnoses in Section E.7.

The chapter is organised around RQ1-RQ3: classification performance, qualitative error and evidence-use patterns, operational telemetry, and integrated interpretation of the findings. Chapter 6 presents the conclusions and recommendations.

5.2. RQ1: Quantitative Classification Performance and Interpretation

RQ1 examined whether the Long_Context strategy improved claim-classification performance relative to the RAG-based strategy for SEC 10-K cross-section inconsistency detection under the controlled same-model design. Table 5.1 presents the headline comparison derived from Appendix E, Section E.6, Table E.6. The two strategies performed closely: the RAG-based strategy classified 42 of 48 claim-cases correctly, compared with 41 of 48 under Long_Context.

Table 5.1: Overall Claim-Classification Performance by Strategy

Context-delivery strategy	Classified correctly	Total claim-cases	Accuracy	Macro F1
Long_Context	41	48	85.42%	84.68%
RAG-based	42	48	87.50%	87.28%

Table 5.2 presents the label-level comparison derived from Appendix E, Section E.6, Tables E.7-E.9. Both strategies correctly classified all 16 true Contradicted claim-cases, producing 100.00% recall for that label. The weaker label-level result was Insufficient Evidence: Long_Context achieved 56.25% recall and 72.00% F1, while RAG-based achieved 68.75% recall and 78.57% F1. For Supported cases, Long_Context classified all 16 correctly, while RAG-based classified 15 of 16 correctly.

Table 5.2: Label-Level Claim-Classification Performance by Strategy

Context-delivery strategy	Expected label	Correct / Total	Precision	Recall	F1
Long_Context	Supported	16 / 16	100.00%	100.00%	100.00%
Long_Context	Contradicted	16 / 16	69.57%	100.00%	82.05%
Long_Context	Insufficient Evidence	9 / 16	100.00%	56.25%	72.00%
RAG-based	Supported	15 / 16	100.00%	93.75%	96.77%
RAG-based	Contradicted	16 / 16	76.19%	100.00%	86.49%
RAG-based	Insufficient Evidence	11 / 16	91.67%	68.75%	78.57%

The RQ1 finding is narrow but important. The RAG-based strategy achieved an advantage of 2.08 percentage points in accuracy and 2.60 percentage points in macro F1, but this small gap within a bounded same-model design does not imply general superiority for the RAG-based strategy. Instead, the finding directly addresses the research gap established in Sections 1.5 and 2.10: full cleaned filing access under Long_Context did not produce superior claim classification for SEC 10-K cross-section inconsistency detection. This aligns with the literature distinction in Sections 2.3-2.4 between nominal context capacity and effective evidence use. Expanded context capacity made distributed filing evidence available, but availability alone did not guarantee correct application across the Supported, Contradicted and Insufficient Evidence boundaries. The main quantitative weakness was lower recall and F1 for Insufficient Evidence, which is examined qualitatively in Section 5.3.

5.3. RQ2: Qualitative Error and Evidence-Use Patterns

RQ2 examined how incorrect outputs used, missed or over-interpreted filing evidence across both context-delivery strategies.

Overall label-direction view: across the 96 completed calls, there were 13 misclassified outputs across 8 distinct claim-cases. Twelve involved cases whose expected label was Insufficient Evidence but which were predicted as Contradicted; the remaining error was a Supported case predicted as Insufficient Evidence. The main error direction was therefore clear: most misclassifications involved over-treating Insufficient Evidence as direct contradiction rather than missed contradiction or random label noise. Appendix E, Section E.6, Table E.9 records the confusion directions, while Appendix E, Section E.7, Table E.12d summarises the 13 outputs across 8 affected claim-cases.

Strategy-level view: The Long_Context strategy produced 7 errors, all involving Insufficient Evidence cases predicted as Contradicted. No Supported or true Contradicted case was misclassified under Long_Context. These errors comprised 1 missing-cross-section-support case, 3 unsupported causal cases and 3 unsupported forward-looking/external cases. The RAG-based strategy produced 6 errors: 5 Insufficient Evidence cases predicted as Contradicted and 1 Supported case predicted as Insufficient Evidence. These comprised 3 unsupported causal cases, 2 unsupported forward-looking/external cases and 1 risk-to-MD&A linkage case. The shared pattern was over-calling insufficiency as contradiction, while the RAG-based strategy also produced 1 Supported-to-Insufficient Evidence error. Appendix E, Section E.7, Table E.12b records this claim-case type pattern.

Claim-case type concentration: unsupported causal and unsupported forward-looking/external claims accounted for 11 of the 13 errors despite representing only 11 of the 48 locked claim-cases. Errors were therefore concentrated rather than evenly distributed across the claim-case type distribution recorded in Appendix C, Table C.5a. This shows that the model struggled most with causal attribution, forward-looking or external assertions, and metric-specific sufficiency, rather than failing broadly across all claim types. Appendix E, Section E.7, Table E.12c provides the error-concentration cross-check.

Claim-case view: 5 claim-cases failed under both context-delivery strategies, producing 10 of the 13 misclassified outputs. These shared failures indicate a common evidential-sufficiency boundary problem rather than a retrieval-only defect. The remaining 3 failures were Apple F001_C08 and Verizon F005_C08 under Long_Context, and Amazon F002_C03 under the RAG-based strategy. These show distinct failure patterns: Long_Context did not eliminate metric-specific or forward-looking sufficiency failures despite full-filing access, while the RAG-based Amazon case showed retrieval incompleteness where a required cross-section evidence component was missing. Appendix E, Section E.7, Table E.12d summarises this distribution, while Tables E.15-E.16 retain the detailed case-level diagnoses.

Comparative synthesis: error quality was more similar than different across the two context-delivery strategies. Both strategies mainly failed at the boundary between Insufficient Evidence and Contradicted, showing that the core weakness was evidential-sufficiency judgement rather than evidence access alone, contradiction detection itself or retrieval failure alone. The model generally recognised direct contradiction, but struggled where a claim was related to disclosed material without being directly established by it. This explains why many errors occurred for causal, forward-looking, external and metric-specific assertions. For evidence-grounded SEC 10-K reasoning, related disclosure is not enough; the supplied

evidence must satisfy the label definitions for support or contradiction. Appendix E, Section E.7, Tables E.15-E.16 show repeated patterns where adjacent financial measures, alternative disclosed drivers, cautionary language, future outlooks or incomplete cross-section evidence were treated as sufficient for contradiction even though the label definitions and scoring rules required Insufficient Evidence.

The single-strategy failures add nuance. Apple F001_C08 and Verizon F005_C08 were misclassified only under Long_Context, showing that full-filing access did not by itself prevent metric-specific or forward-looking sufficiency mistakes. Amazon F002_C03 was misclassified only under the RAG-based strategy, showing that retrieval can fail when one side of a cross-section evidence relationship is missing. Appendix E, Section E.7, Table E.12d summarises the shared and single-strategy distribution.

All 13 misclassified outputs reported High confidence, so the confidence field did not identify the observed failures. The RQ2 interpretation is therefore that context-delivery strategy affects what evidence is supplied, but the harder reasoning requirement is applying that evidence with label-boundary discipline. This connects to Sections 2.4 and 2.8, where the literature review distinguishes expanded context capacity from reliable evidence-grounded reasoning.

5.4. RQ3: Operational Telemetry and Trade-Offs

RQ3 examined the operational trade-off between the two context-delivery strategies. Table 5.3 summarises the telemetry comparison derived from Appendix E, Section E.5, Table E.4. Long_Context used 8,582,196 input tokens and 25,257 output tokens, compared with 443,289 input tokens and 20,111 output tokens under the RAG-based strategy. The main operational difference was therefore input-context scale rather than output length.

Table 5.3: Operational Telemetry by Context-Delivery Strategy

Context-delivery strategy	Input tokens	Output tokens	Telemetry-estimated model-call cost	Mean latency	Median latency
Long_Context	8,582,196	25,257	\$43.668690	10.36s	9.19s
RAG-based	443,289	20,111	\$2.819775	6.23s	5.34s

In proportional terms, Long_Context used approximately 19.36x more input tokens and 1.26x more output tokens than the RAG-based strategy. It also incurred approximately 15.49x higher telemetry-estimated model-call cost, with mean latency 1.66x higher and median latency 1.72x higher. These costs are telemetry-estimated model-call costs calculated from logged token counts, not provider-billed charges.

The RQ3 finding is therefore clear: Long_Context imposed substantially higher token use, telemetry-estimated model-call cost and latency without producing the classification

advantage from the larger context. Full-filing access carried a clear operational burden, but the extra context volume did not translate into better classification performance in this experiment. This aligns with Section 2.9, where the literature review justified token use, telemetry-estimated model-call cost and latency as operational dependent measures alongside classification quality.

RQ3 therefore supports a bounded interpretation of context-delivery choice. A larger context window may reduce dependence on retrieval, but it also increases token use, cost and latency. Conversely, RAG-based context delivery was operationally more efficient, but its value remains dependent on retrieval completeness. The practical trade-off is not “more context is better”; it is whether additional context improves evidence-grounded decisions enough to justify its operational cost.

5.5. Interpretation in Relation to Long-Context and RAG Literature

The combined RQ1-RQ3 findings support the literature-review distinction between nominal context capacity and effective evidence use, particularly Sections 2.3-2.4 on long-context limitations and Sections 2.5-2.6 on RAG trade-offs. The Long_Context findings provide a domain-specific example of this distinction: full SEC 10-K filings were available to the model, yet errors still occurred at the boundary between Insufficient Evidence and Contradicted. This supports a cautious interpretation of long-context capability. Complete filing access can remove retrieval omission as an explanation, but it does not guarantee consistent evidence identification, integration or label-boundary application.

The RAG-based findings reflect the retrieval trade-offs discussed in Sections 2.5-2.6 and the operational-measure rationale set out in Section 2.9. Retrieval substantially reduced the amount of filing text supplied to the model while preserving slightly higher overall accuracy and macro F1. This supports the practical role of the RAG-based strategy as a context-selection mechanism rather than as a superior reasoning method. However, Appendix E, Section E.7, Table E.16 shows that Amazon F002_C03 failed because the frozen retrieved context contained AWS/AI material but omitted the required MD&A evidence component. The retrieved context was therefore related to the claim but evidentially incomplete.

The balanced interpretation is that the Long_Context strategy reduced retrieval-completeness risk by supplying the full cleaned filing, but it did not guarantee evidence-use discipline at the boundary between Insufficient Evidence and Contradicted. The RAG-based strategy substantially reduced input-context size, telemetry-estimated model-call cost and latency, but remained dependent on retrieval completeness. Only one RAG-based error was directly established as retrieval incompleteness; the other RAG-based errors primarily reflected the same evidential-sufficiency boundary that also affected Long_Context. In relation to the

literature reviewed in Sections 2.3-2.6 and 2.9, the experiment therefore does not show that either context-delivery strategy is generally superior. It shows that both strategies expose different trade-offs: Long_Context reduces retrieval omission but can still fail in evidence-use discipline, while RAG-based improves context efficiency but can omit necessary cross-section evidence.

5.6. Implications for Evidence-Grounded SEC 10-K Reasoning

The findings have specific implications for SEC 10-K reasoning because financial disclosures distribute relevant evidence across sections including Business, Risk Factors, MD&A, financial statements and notes. This links to Sections 2.7-2.8, where financial-document reasoning is framed as a claim-specific evidence-grounding problem rather than a basic text-retrieval task. A model must therefore decide whether combined cross-section evidence directly supports, directly contradicts or remains insufficient for the specific claim being evaluated.

The qualitative error pattern shows why this distinction matters. Multiple failures involved unsupported causal, forward-looking, external or metric-specific assertions. Filings may report a financial movement alongside an initiative, risk, transaction or market condition without establishing the precise causal, predictive or quantitative claim being tested. Treating that absence of sufficient support as contradiction can overstate what the filing proves.

Appendix E, Section E.7, Table E.16 illustrates this point at case level. Apple F001_C08 shows that related financial measures are not interchangeable, because iPhone net sales did not establish an exact unit-sales count. Amazon F002_C03 shows the cross-section retrieval risk, because the retrieved context supplied AWS/AI material but omitted the required MD&A evidence component.

These failure modes would have different implications in analytical use. False Contradicted outputs could overstate filing inconsistency, while false Insufficient Evidence outputs could miss a supported cross-section relationship. These implications remain bounded because the experiment did not measure real-world analyst decisions or financial harm. For consequential financial-document review, model outputs would require evidence traceability and human review, and should not be treated as investment advice, audit conclusions, legal determinations or financial recommendations, consistent with the ethics and data-handling boundary in Appendix F, Section F.2.

5.7. Validity and Apparatus Boundaries

The findings must be interpreted within the limits of the controlled apparatus defined in Chapter 3 and implemented in Chapter 4. The study is not a general benchmark of all long-context models, RAG configurations or financial-document reasoning tasks. Its conclusions are

bounded to the experimental scope, controls and validity conditions defined in Sections 1.6, 3.2 and 3.11.

Internal comparability was supported by the controlled same-model design: the selected model, claim-cases, label definitions, prompt structure, output schema, scoring procedure and telemetry-capture procedure were held constant, while context-delivery strategy was varied. These controls are defined in Sections 3.2 and 3.11 and operationalised through the Chapter 4 execution pipeline, so they are not repeated in detail here.

Construct and conclusion validity are also bounded. The claim-cases and qualitative diagnoses were researcher-defined, with no independent dual-coding exercise. The RAG-based strategy classified one additional claim-case correctly, but this difference is descriptive and should not be read as statistical significance. Filing-level error totals should not be interpreted as company, sector or filing-size effects; the observed pattern must be read alongside claim-case type composition and the bounded dataset design.

The apparatus was also limited to one Long_Context prompt design and one frozen RAG configuration using fixed chunking, overlap and top-k retrieval. Alternative chunking, reranking, hybrid routing, concurrency, throughput, rate-limit behaviour and production-scale load were not evaluated. These boundaries do not invalidate the comparison, but they confine the conclusions to the locked experimental apparatus and retained evidence base.

5.8. Chapter Summary

This chapter presented the main-run analysis for RQ1-RQ3, covering claim-classification performance, qualitative error and evidence-use patterns, operational telemetry and their interpretation in relation to the literature. It also identified the study's validity boundaries and the implications of the retained evidence base.

Chapter 6 synthesises these findings into the dissertation's response to the research aim, key contributions, recommendations, reflective evaluation and overall conclusion.

Chapter 6: Recommendations and Conclusions

6.1. Synthesis of the Main Findings

Chapter 5 findings establish that the Long_Context strategy did not provide a measurable classification advantage over the RAG-based strategy in this bounded task-specific experiment. The two strategies performed closely, with the RAG-based strategy classifying one additional claim-case correctly and achieving marginally higher accuracy and macro F1. Both strategies identified all true Contradicted claim-cases, while Insufficient Evidence remained the weaker label for both strategies, especially under Long_Context.

The qualitative analysis showed that the central issue was not simple evidence access, but evidential-sufficiency judgement at the boundary between Insufficient Evidence and Contradicted. Most misclassified outputs involved expected Insufficient Evidence cases being over-treated as Contradicted, particularly where causal, forward-looking, external or metric-specific claims were related to filing content but not directly established by it.

The operational findings showed that Long_Context carried substantially higher input-token use, telemetry-estimated model-call cost and latency without producing a classification advantage.

Taken together, these findings indicate that expanded context capacity did not, by itself, deliver more reliable evidence-grounded SEC 10-K reasoning in the evaluated setup.

6.2. Key Contributions

This dissertation makes a bounded *empirical contribution* by providing a controlled same-model comparison of the Long_Context and RAG-based strategies for evidence-grounded SEC 10-K cross-section inconsistency detection. It shows, within this experimental setup, that full-filing access did not provide a measurable classification advantage over retrieval-mediated context delivery.

Methodologically, the dissertation contributes a reproducible controlled evaluation structure based on locked filings, balanced evidence-anchored claim-cases, hidden expected labels, consistent task instructions, a fixed output schema, retained execution records, telemetry capture, exact label scoring and appendix-level audit evidence. This structure isolated context-delivery strategy as the independent variable.

Analytically, the dissertation distinguishes context completeness from evidence-use discipline. The case-level analysis separated one RAG-specific retrieval-completeness failure from the

dominant error pattern shared by both strategies: treating insufficient causal, forward-looking, external or metric-specific evidence as direct contradiction. This provides a more precise basis for evaluating evidence-grounded SEC 10-K reasoning than accuracy alone.

6.3. Recommendations for Practice

The findings support four task-specific recommendations for future evidence-grounded SEC 10-K reasoning experiments.

Strategic selection over nominal capacity: systems should not assume that a larger context window will produce better evidence-grounded classification. In this experiment, the Long_Context strategy provided full filing access but no classification advantage, while the RAG-based strategy reduced token use, telemetry-estimated model-call cost and latency but remained dependent on retrieval completeness. Strategy choice should therefore be validated against the task's evidence requirements and operational constraints, rather than inferred from nominal context capacity alone.

Retrieval-completeness checks for RAG-based workflows: RAG-based workflows should include explicit coverage checks for the identifiable evidence components of multi-section claims, rather than assuming that generally relevant retrieved passages are sufficient. In practice, this could involve decomposing a claim into required evidence components, retrieving against those components separately, and escalating to wider-context or human review when required evidence remains missing.

Strict label-boundary enforcement: practical systems should preserve an explicit Insufficient Evidence outcome and require the model to distinguish missing or indirect support from direct contradiction. This is particularly important for causal attribution, forward-looking statements, guarantees and claims involving adjacent but non-equivalent financial measures.

Independent reliability verification: model-reported confidence should not be used as a standalone reliability threshold. All incorrect outputs in this study reported High confidence, although the measure was not calibrated. Human review and evidence-level traceability remain necessary for consequential uses such as investment research, audit support, risk analysis or management-disclosure review.

6.4. Recommendations for Future Research

Future work should test the stability of these findings across repeated executions, additional long-context models, larger SEC 10-K corpora and broader claim-case samples. Repeated executions would quantify output variability, while a more diverse corpus would strengthen generalisability across industries, filing structures and reporting periods.

Expected-label and evidence-anchor development should be reinforced through independent review, dual coding or adjudication, especially at the boundary between Insufficient Evidence and Contradicted, where disclosure may appear contrary without meeting the strict requirement for direct conflict.

Retrieval-focused studies should evaluate whether claim decomposition and component-level coverage checks improve performance on multi-section claims. Section-targeted retrieval, reranking, hybrid retrieval and wider-context escalation should be tested experimentally, alongside systematic variation of chunk sizes, overlap values, retrieval depths and embedding configurations.

Further research should examine whether calibrated confidence, abstention mechanisms or evidence-quality indicators can better identify unreliable outputs than the uncalibrated confidence field used here. Evaluation should extend beyond classification accuracy to evidence completeness, citation correctness, output stability, operational throughput and human-review requirements.

Together, these extensions would provide a stronger basis for determining when Long_Context, RAG-based or hybrid strategies are most appropriate for evidence-grounded SEC 10-K reasoning.

6.5. Reflection on the Research Process

The *Type 5* experimental approach was appropriate because the dissertation required implementation, controlled execution and measurable comparison rather than conceptual analysis alone. Developing the filing register, locked claim-cases, scoring key, prompts, retrieved contexts and audit artefacts in advance helped preserve the intended controlled same-model comparison and reduced the risk of retrospective adjustment after observing results.

A major methodological lesson was the importance of separating evidence preparation, execution, scoring and interpretation. Withholding expected labels from model-facing inputs, freezing the RAG-based contexts before scored execution, and retaining raw outputs, telemetry and scoring records supported traceability. The later case-level error analysis also showed the value of examining individual evidence relationships rather than relying only on aggregate accuracy and macro F1.

The process demonstrated that reproducibility in API-based LLM experiments depends on transparent control and evidence retention rather than claims of exact output determinism. Provider-side behaviour could not be fully controlled, but consistent request settings, retained model identifiers, locked inputs and retained execution evidence supported a fair comparison within the implemented setting.

6.6. Overall Conclusion

This dissertation aimed to determine whether the Long_Context strategy, using the selected GPT-5.5 API model, provided measurable advantages over the RAG-based strategy for SEC 10-K cross-section inconsistency detection. The findings do not support such an advantage within the bounded experimental setup evaluated here. Instead, they show that making more filing text available to the selected GPT-5.5 API model may reduce retrieval dependence, but it does not guarantee that the model will integrate evidence correctly or apply label boundaries reliably.

Overall, neither the Long_Context strategy nor the RAG-based strategy was sufficient by itself to guarantee accurate evidence-grounded SEC 10-K reasoning. Effective use of context-delivery strategies requires claim-specific evidence integration, disciplined separation of Insufficient Evidence from direct contradiction, and adequate coverage of all filing components needed to assess a claim. Within its bounded scope, the dissertation provides empirical, methodological and analytical evidence for designing future evaluations of long-document financial reasoning systems.

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Appendices

The appendices and supporting artefacts are submitted in the accompanying `Dissertation_Submission_Artefacts.zip` package. Paths in the table below are relative to that package root.

The dissertation report is the primary reading route; the appendix files provide readable supporting material, while the evidence, codebase and audit folders provide reproducibility and verification material where required.

Table A.1: Appendix and Submission Artefact Register

Appendix / Artefact	Submitted location	Purpose
Appendix A1	02_Appendices/Appendix_A1_Structured_Literature_Review_Protocol_Final.pdf	Structured literature review protocol, including review scope, PICOC framing, search themes, screening criteria, extraction fields and synthesis approach.
Appendix A2	02_Appendices/Appendix_A2_Citation_Control_Matrix_Final.xlsx	Citation-control matrix recording validated sources, source roles, citation boundaries and supporting pin-points.
Appendix B	02_Appendices/Appendix_B_Methodology_and_Experiment_Protocol_Final.pdf	Methodology and experiment protocol, including parameter locks, execution procedure, retry rules, logging fields and reproducibility controls.
Appendix C	02_Appendices/Appendix_C_Dataset_and_Claim_Case_Register_Final.pdf	Dataset and claim-case register, including filing metadata, claim-cases, expected labels, evidence anchors, rationales, scoring rules and claim-case type distribution.
Appendix D	02_Appendices/Appendix_D_Prompt_Templates_Output_Schema_and_RAG_Configuration_Final.pdf	Prompt templates, shared output schema, RAG configuration and locked retrieved-context structure.
Appendix E	02_Appendices/Appendix_E_Experiment_Run_Logs_Cost_Latency_and_Results_Evidence_Final.pdf	Main-run evidence summary, including outputs, telemetry, metrics, confusion matrices and qualitative error evidence.
Appendix F	02_Appendices/Appendix_F_Ethics_Reproducibility_and_Audit_Material_Final.pdf	Ethics, reproducibility, pilot/pre-main-run validation boundary, risk register, audit trail and final artefact-control material.
Main-run Evidence Pack	03_Main_Run_Evidence/Appendix_E_Evidence_Pack	Machine-readable main-run evidence, including validation reports, prompt hashes, prompt-size summaries, telemetry, scored predictions, metrics, confusion matrices and related evidence files.

Derived Analysis Workbook	03_Main_Run_Evidence/Derived_Analysis/Main_Run_Results_Extraction_and_Analysis_Workbook_v1.xlsx	Workbook used to derive and verify final metrics, telemetry summaries and analysis tables from retained main-run evidence.
Locked Source Evidence	03_Main_Run_Evidence/Locked_Source_Zips/	Locked source ZIPs preserving full-live outputs, scoring outputs and Appendix E evidence-pack material.
Codebase	04_Codebase/CS958-ROUS04-Codebase.zip	Final submitted codebase and reproducibility environment, excluding secrets and local credentials.
Audit and Control Material	05_Audit_and_Control/Final_Submission_Artefact_List.pdf	Final artefact list, pilot validation summary, pilot workbook, pre-main-run validation reports and locked pre-main-run source ZIPs.
Combined Appendix Pack	02_Appendices/Appendices_A_to_F_Pack_Final.zip	Convenience package containing the final Appendix A-F documents. Individual appendix files are also provided for direct access.